

WHAT'S THE MATTER WITH MATTER?

In the first moments after the big bang, tiny particles of energy from the explosion converted into particles of matter—the substance of which things are made. Much later, when the universe was about 300,000 years old, the first atoms began to form from this matter. Made of hydrogen and helium gas, these atoms went on to create the other chemical elements that exist today. Earth and eventually everything on it came from these elements, yet they make up less than five percent of the universe. With only a tiny part accounted for, it wasn't the end of the matter...

MYSTERIOUS UNIVERSE

The other 96 percent of the universe is something of a mystery. It appears that it is made up of two types of hidden substances—unknown matter called dark matter and a mysterious force called dark energy. Although it is not possible to directly detect them, scientists believe they exist because of their effect on objects that we can see.

DARK FORCES

The invisible dark matter does not release any detectable energy, but its gravity has an effect on its surroundings. Let's start in space, where galaxies are arranged in clusters. Although these galaxies are visible to astronomers, there are movements of galaxies within clusters that can be explained only by the

gravity of an unseen material pulling on them as they move in the universe. Known as dark matter, this material makes up about 23 percent of the entire universe. Do the math—that's 27 percent explained. So, what about the rest? A whopping 73 percent of the universe is believed to be an unknown form of energy. In 1998, astronomers studying supernovae (exploding stars) showed how they could be used to determine the speed of the universe's expansion. It was thought that the expansion rate was slowing down, but the supernovae suggested it was actually speeding up. Astronomers discovered a repulsive force pushing the universe apart that modifies the gravity of ordinary matter pulling it together. Enter dark energy. While the gravitational force of dark matter helps pull the universe together, the repulsive force of dark energy makes the universe expand more rapidly.

MATTER OF FACT

Theories are being developed all the time that attempt to offer a different version of the universe with different, or no, roles for dark matter and dark energy. However, many of these theories have been overturned by scientists. New ideas are usually dismissed, as they do not stand up to further research and investigation. Dark matter and dark energy remain the accepted explanations, no matter what.